

Statistical Analysis II - Hypothesis Testing, A/B Testing, Performance Metrics

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Hypothesis Testing

Motivation

Motivation

Why do we need hypothesis testing?

- ▶ It is a systematic method to answer questions, such as whether two sets of data are equivalent.
- ▶ Given the randomness, it provides a probabilistic statement for the likelihood of observing the data under a hypothesis.
- ▶ With the qualities above, it is a powerful tool in business decision making process, e.g. A/B tests.

Fundamental concepts

Null and alternative hypothesis

Null hypothesis denoted by H_0

- ▶ It is usually the conservative hypothesis, or what we want to prove to be wrong.
- ▶ We assume the null hypothesis and use data observations to decide whether to reject the null hypothesis or not.

Alternative hypothesis denoted by H_1 or H_A

- ▶ This is usually the main purpose of the study, or what we want to prove to be correct.
- ▶ There is no overlap between H_0 and H_1 , and in most cases they should be complementary of each other, meaning $H_0 \cup H_1$ exhausts the whole space.

Food for thought

- ▶ A pharmaceutical company is working on a new drug for cancer treatment. What do you think the null and alternative hypotheses should be?
- ▶ Federal Reserve is trying to determine whether the stimulus plan has an impact on recipient's financial well-being. What is its null and alternative hypothesis?

One-sided testing example

$$H_0: \mu \leq 65 \quad \text{vs} \quad H_1: \mu > 65$$

$$\underline{X \sim \mathcal{N}(\mu, 25) - \text{assumed}}$$

$$X: \text{car speeds}$$
$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

observed

The speed limit imposed on highway is 65 MPH. We want to test whether the average car speeds on highway is above 65 MPH.

$$P(\bar{X} > \bar{x})$$

$$\bar{X} = \frac{1}{n} \sum X_i \quad - \text{r.v.} \quad \text{iid.} \quad \mathcal{N}(\mu, 25)$$

$$\bar{x} = \frac{1}{n} \sum X_i \quad - \text{observed}$$

$$\mathbb{E} X_i = \mu$$
$$\mathbb{E} \bar{X} = \mu$$

$$= \mathbb{E} \left(\frac{1}{n} \sum X_i \right)$$
$$= \frac{1}{n} \sum \mathbb{E} X_i$$
$$= \frac{1}{n} \times n \times \mu$$

Test statistic

$$P(\bar{X} > \bar{x}) = P\left(\frac{\bar{X} - \mu}{s/\sqrt{n}} > \frac{\bar{x} - 65}{s/\sqrt{n}}\right) = \boxed{P(Z > \frac{\bar{x} - 65}{s/\sqrt{n}})}$$

$$X_i \sim N(\mu, 25) \quad \sim N(0, 1) = Z$$

p-value

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \sim N\left(\mu, \frac{25}{n}\right)$$

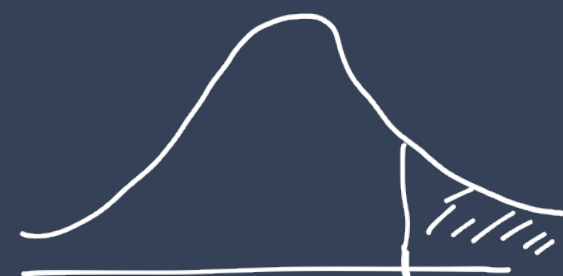
$$\text{var}(\bar{X}) = \text{var}\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \text{var}\left(\sum_{i=1}^n X_i\right)$$

$$= \frac{1}{n^2} \times n \times \text{var}(X_i)$$

$$= \frac{1}{n} \times 25$$

$$\boxed{\frac{\bar{x} - 65}{s/\sqrt{n}}} \text{ - test statistic}$$

z-score



P-value

Naturally, if $P(\bar{X} > \bar{x})$ is small, we have enough evidence to believe the alternative hypothesis is true. However, we now have a new problem. How large is considered large for $P(\bar{X} > \bar{x})$?

- ▶ This is known as the p-value.
- ▶ The interpretation for p-value is: under the null hypothesis, p-value is the probability of observing data as or more extreme as what is at hand.
- ▶ The smaller the p-value, the more unlikely null hypothesis is to hold true.

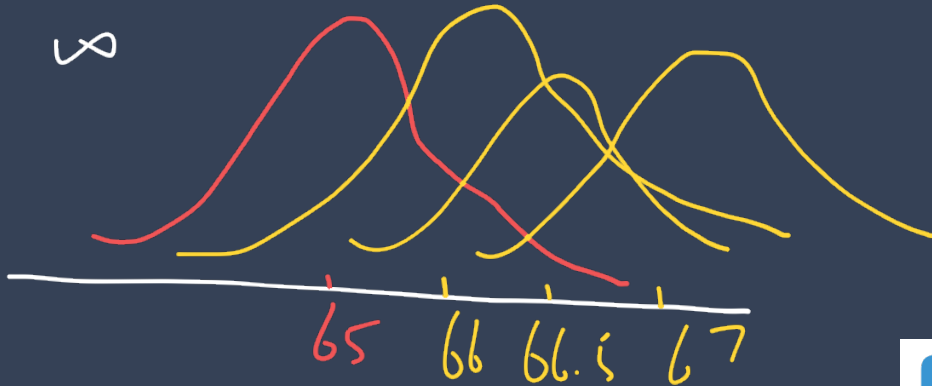
Food for thought

$$H_0: \mu = 65 \quad \text{vs} \quad H_1: \mu > 65$$

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\sim D(\mu)$$

- ▶ Why is the premise “under the null hypothesis” important in the definition of P-value?
- ▶ How many different distributions are there under the null hypothesis?
- ▶ How many are there under the alternative hypothesis?

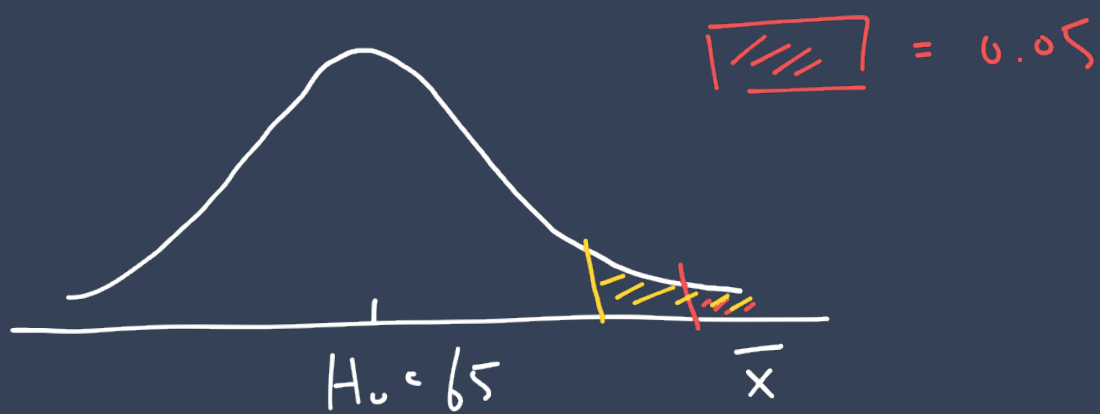


Significance level

- ▶ Popular thresholds for p-value are 0.01, 0.05 and 0.1, depending on the application.
- ▶ For example, in medical studies, where the consequence of rejecting the null hypothesis is more costly, 0.01 is used more often. In day to day business decisions, 0.05 or even 0.1 are common choices.
- ▶ The threshold is also known as significance level, and is often denoted by α .
- ▶ Significance level is predetermined before the hypothesis test.

Given a significance level α , if p-value is less than the significance level, we say there is enough evidence to reject the null hypothesis. Otherwise, we say there is not enough evidence to reject the null hypothesis. Significance level is also known as the tolerance level for Type-I error.

Food for thought



Why is it the case that when p-value is less than significance level, we reject the null hypothesis?

$$0.05 \times 0.05 = 0.0025$$

Type-I and Type-II error

- ▶ Type-I error - When the null hypothesis H_0 is true, the probability that we reject the null hypothesis H_0 .
- ▶ Type-II error - When the alternative hypothesis H_1 is true, the probability that we fail to reject the null hypothesis H_0 .

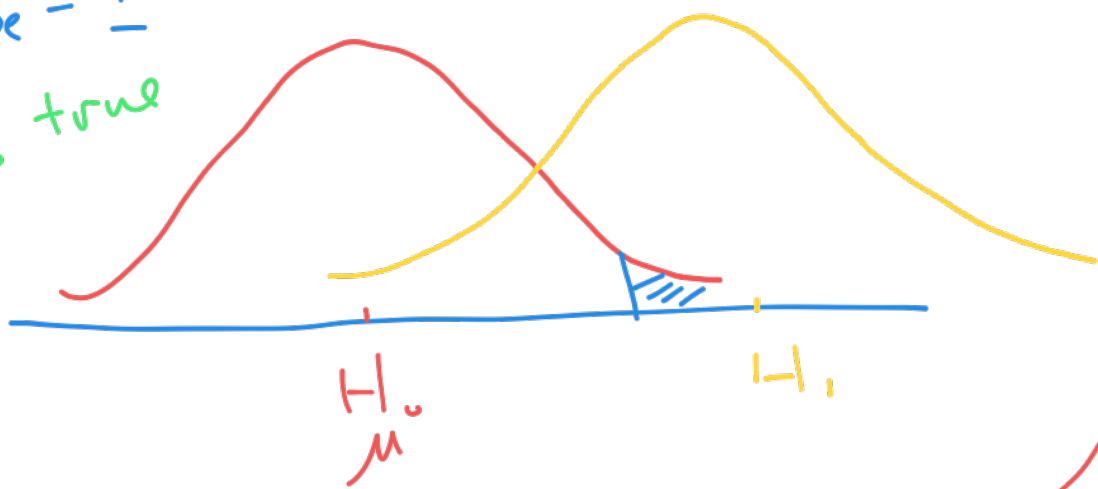
Summarized in the a table format ¹.

Type I and Type II Error		
Null hypothesis is...	True	False
Rejected	Type I error False positive Probability = α	Correct decision True positive Probability = $1 - \beta$
Not rejected	Correct decision True negative Probability = $1 - \alpha$	Type II error False negative Probability = β



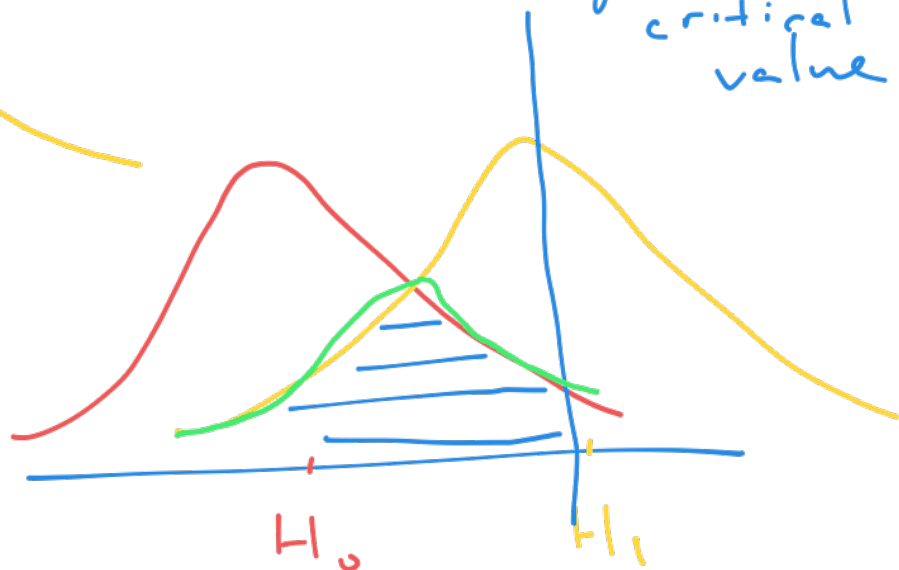
¹Plot credit to scribbr.com

Type - I
 H_0 true

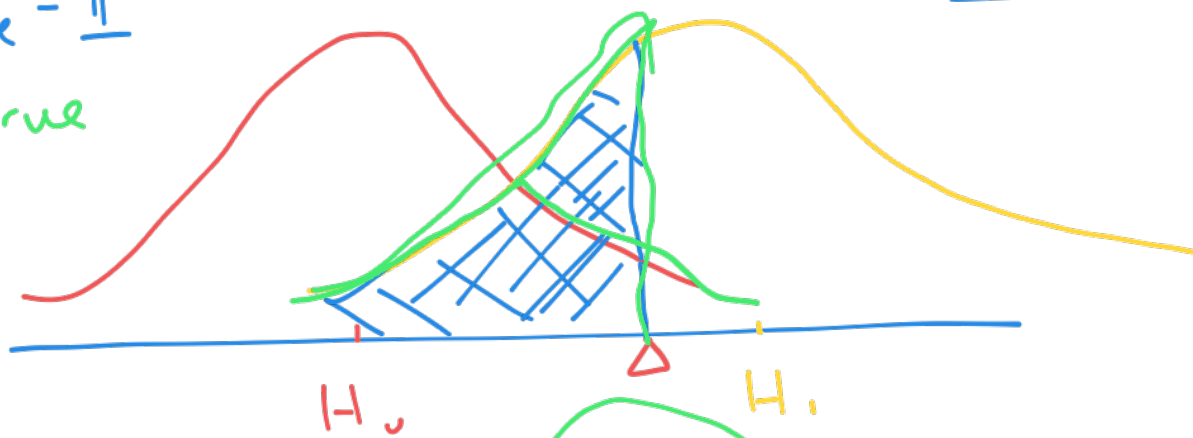


$\bar{X}$

significant
critical
value



Type - II
 H_1 true



$$\mu = 70$$

$$\mu = \bar{X}$$

$$\mu = 80$$



Food for thought

What is type-I error equal to?

- ▶ p-value
- ▶ significance level

Can type-II error be easily quantified? Why don't we control type-II error?

Confidence interval

Two-sided testing example

Human heart rate is believed to average at 80. A new supplemental drug has been introduced, and the manufacturer wants to understand whether the supplement changes patients' heart rates.

Two-sided testing example

Food for thought

Why do we need to consider the more extreme one between $P(\bar{X} > \bar{x})$ and $P(\bar{X} < \bar{x})$ in the two-sided case?

Z-test

Z-test

- ▶ The examples we went through earlier are known as Z-tests. They work under the assumption that the random variable is normally distributed. (“Z” is often associated with the normal distribution as in “z-statistic” and “z-table”).
- ▶ Z-tests assume known variance, as demonstrated by the two examples that we went through. What if the variance of the data underlying distribution is not known? Is there a remedy so that we can carry out the hypothesis test?

Proportion test

Small sample proportion test

- ▶ Proportion test is used for testing the success rate parameter for a binominal distribution, which consists of binary events.
- ▶ For example, we have the 5 independent trials of a coin toss, and they are all $\{HHTHH\}$, and we want to test whether the coin is fair.
- ▶ In other words, we want to test whether $H_0 : p = 1/2$ versus $H_1 : p \neq 1/2$, which is equivalent to $H_0 : \underline{P(H) = 1/2}$ versus $H_1 : P(H) \neq 1/2$.

Small sample proportion test

$$H_0: p = \frac{1}{2} \quad \text{vs} \quad H_1: p \neq \frac{1}{2} \quad \alpha = 0.05 \quad 5 \text{ trials}$$

$$E(X) = 2.5 \quad X = \# \text{ of heads}$$

extreme values > 2.5 or < 2.5 , we reject the H_0

X	0	1	2	3	4	5
P(X)	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

$$P(X=0) + P(X=5) = \frac{1}{32} + \frac{1}{32} = \frac{1}{16} = 0.0625$$

reject if $X=0$ or $X=5$ > 0.05

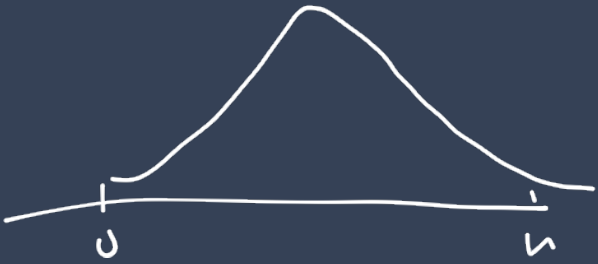
X	0	1	5	6
P(X)	$\frac{1}{64}$			$\frac{1}{64}$

$$P(X=0) + P(X=6) = \frac{1}{32} < 0.05$$

Large sample proportion test

binomial $\approx$ Normal $n > 30$
 n large

- ▶ One can imagine as sample size increases, the computation becomes intractable.
- ▶ Say we have n observations, and we want to test $H_0 : p = p_0$ versus $H_1 : p \neq p_0$.
- ▶ If $0 < \frac{np_0 - 3\sqrt{np_0(1-p_0)}}{np_0 + 3\sqrt{np_0(1-p_0)}} < n$, then define k as the number of success trials we observe,



$$z = \frac{k - np_0}{\sqrt{np_0(1 - p_0)}} \quad (1)$$

can be used as a z-statistic.

Food for thought

Why do you think the large sample proportion test can be well approximated with z-test?

Large sample proportion test

A drug manufacturer has been testing a new drug with 10,000 participants and observed 1615 participants with side effects. They think the new drug has a side effect rate of less than 0.15, and want to verify their hypothesis at 0.01 significance level.

T-test

T-test versus Z-test

- ▶ T-tests work similarly to Z-tests, except that the t-statistic follows a different distribution than normal, known as the (Student's) T-distribution.
- ▶ When do we use T-test instead of Z-test? - When (1) variance of the underlying data distribution is not known and (2) the number of observations is small.

T-test versus Z-test

$$\mathbb{E} \left((X - \mathbb{E} X)^2 \right)$$

- ▶ When variance of the underlying data distribution is not known, additional standard deviation estimator is needed in place of the known standard deviation, thus making the test statistic no longer normally distributed.
- ▶ When large number of observations are available, central limit theorem is usually in effect, which alleviates the fact that the test statistic is not exactly normally distributed, but is normally distributed in limit.

T-test

A key fact that we know is that for $X \sim \mathcal{N}(\mu, \sigma^2)$, with some unknown σ^2 ,

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \quad (2)$$

follows a T_{n-1} distribution, where

$$S = \sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)} \quad (3)$$

When degrees of freedom is large, e.g. $> 1,000$, T-distribution can be well approximated by a standard normal distribution.

T-test example

Say we observe the speeds of 10 cars on the highway, and they are $\{80, 77, 60, 72, 63, 69, 74, 68, 63, 79\}$. We know that car speeds on the highway follow a normal distribution. How can we test whether the mean car speed on the highway μ exceeds 65 with significance level 0.05, i.e. $H_0 : \mu = 65$ versus $H_1 : \mu > 65$?

T-test example - test statistic

T-test example - p-value

T-test example - confidence interval

A/B Testing

Motivation

Motivation

- ▶ What is A/B testing? - Given version A and B, A/B testing helps answer the question: which version is better.
- ▶ Why do we need A/B testing? - Successful business decisions cannot rely on intuition entirely. A/B testing offers a data driven method to determine whether a change is good.

Examples

Modern day internet companies owe much of their success to massive number of A/B tests. They are used to answer many day to day questions, such as

- ▶ Is an advertisement banner better at catching user attention with a bold or regular font?
- ▶ Do users prefer status bar at the top or on the side? If it is on the side, is it left or right?
- ▶ A new video recommendation algorithm has been developed, how do we determine that it is better than the current one?

Two-sample proportion test

Two-sample proportion test

$$p_x = P(X_i = 1)$$

- ▶ Two-sample proportion test is a hypothesis test, which aims to answer the question whether two samples of binary events have equal positive proportions.
- ▶ Given a sample of $\{X_1, X_2, \dots, X_n\}$ and $\{Y_1, Y_2, \dots, Y_m\}$, each is generated from Bernoulli distributions with parameters p_X and p_Y . We are interested in testing

$$H_0 : p_X = p_Y \text{ versus } H_1 : p_X \neq p_Y \quad (4)$$

Two-sample proportion test

$$H_0: p_x = p_y$$

$$H_1: p_x \neq p_y$$

$$H_0: \underbrace{p_x - p_y}_{\text{circled}} = 0$$

$$H_1: p_x - p_y \neq 0$$

Let x and y be the number of successes in the two samples of Bernoulli trials, and define $p_e = (x + y)/(n + m)$, then we know

$$z = \frac{\frac{x}{n} - \frac{y}{m} - 0}{\sqrt{p_e(1 - p_e) \left(\frac{1}{n} + \frac{1}{m} \right)}} \quad \text{assume } H_0 \quad (5)$$

follows a standard normal distribution. $\frac{1}{n} p_e(1 - p_e) + \frac{1}{m} p_e(1 - p_e)$
 $\frac{1}{n} \frac{x}{n} \left(1 - \frac{x}{n} \right) + \frac{1}{m} \frac{y}{m} \left(1 - \frac{y}{m} \right)$

Two-sample proportion test example

Two advertisement campaigns are being tested out in two groups of 10,000 customers each. For the first campaign, 139 customers clicked on the banner. For the second campaign, 147 customers clicked on the banner. Are the response rates for the two campaigns similar at 0.05 significance level?

Two-sample T-test

Two-sample T-test

- ▶ Two-sample T-test is a hypothesis test, which aims to answer the question whether two samples have equal means.
- ▶ Given a sample of $\{X_1, X_2, \dots, X_n\}$ and $\{Y_1, Y_2, \dots, Y_m\}$, each generated from normal distributions with means μ_X and μ_Y . We are interested in testing

$$H_0 : \mu_X = \mu_Y \text{ versus } H_1 : \mu_X \neq \mu_Y \quad (6)$$

Two-sample T-test with unknown and unequal variances

Under the assumption $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^m$ are normally distributed with means μ_X and μ_Y and unequal standard deviations, we define test statistic

$$W = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}, \quad (7)$$

where $\bar{X} = \sum_{i=1}^n X_i/n$ and $\bar{Y} = \sum_{i=1}^m Y_i/m$, $S_X^2 = \sum_{i=1}^n (X_i - \bar{X})^2/(n-1)$ and $S_Y^2 = \sum_{i=1}^m (Y_i - \bar{Y})^2/(m-1)$, let $\hat{\theta} = s_X^2/s_Y^2$ and ν be the nearest integer to the expression

$$\frac{\left(\hat{\theta} + n/m\right)^2}{\frac{\hat{\theta}^2}{n-1} + \frac{(n/m)^2}{m-1}}. \quad (8)$$

Then W has approximately a Student T-distribution with ν degrees of freedom.

Two-sample T-test with unknown and equal variances

Under the assumption $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^m$ are normally distributed with means μ_X and μ_Y and equal standard deviations, the test statistic

$$T = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}}, \quad (9)$$

where the pool variance S_p is defined as

$$\begin{aligned} S_p^2 &= \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^m (Y_i - \bar{Y})^2}{n+m-2} \end{aligned} \quad (10)$$

has a Student T-distribution with $n + m - 2$ degrees of freedom.

Two-sample T-test with known variances

||

$$H_0: \mu_X = \mu_Y$$

$$H_1: \mu_X \neq \mu_Y$$

Under the assumption $\{X_i\}_{i=1}^n$ and $\{Y_i\}_{i=1}^m$ are normally distributed with means μ_X and μ_Y and known standard deviations σ_X^2 and σ_Y^2 , the test statistic

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}, \quad (11)$$

has a standard normal distribution.

A/B test real example

A/B test real example

- ▶ A video sharing company wants to understand whether its new recommendation algorithm is better than the current one.
- ▶ The company has determined that they value “better” by the amount of time users spend on their site.
- ▶ The company decides to use an A/B test to determine whether the new algorithm is better, by exposing 55K randomly selected users to experience A: current recommendation algorithm, and 60K randomly selected users to experience B: new recommendation algorithm.
- ▶ After a week, the company has observed that the customers in A has average site time of $\bar{X} = 58$ minutes with sample standard deviation $s_{\bar{X}}^2 = 13$, whereas customers in B has average site time of $\bar{Y} = 60$ minutes with sample standard deviation $s_{\bar{Y}}^2 = 15$.

A/B test real example

Performance Metrics



Motivation

Motivation

Why do we need performance metrics?

- ▶ To understand how well models are trained.
- ▶ To establish expectation on performance in production.
- ▶ To answer business questions, such as opportunity estimation and gap analysis.

Different tasks have different need for performance metrics. On a high level, they all try to understand the difference between prediction and the target. We will focus on two types of targets:

- ▶ classification
- ▶ continuous

Classification targets

Classification targets

- ▶ For classification targets, we focus on the binary ones, as most multi-class targets can be reduced to binary targets.
- ▶ Binary targets are the ones that take value from the set $y \in \{0, 1\}$, i.e. the true target value is True/False or Positive/Negative.
- ▶ Prediction values for binary classifications are usually either hard labels from the set $\hat{y} \in \{0, 1\}$ or soft labels from the range $\hat{y} \in [0, 1]$.
- ▶ Even though soft labels seem more precise, as they entail a degree of confidence in the score, hard labels eventually need to be derived from soft labels using a threshold in practice.

Binary targets

The immediate metric that comes into mind is the proportion of hard labels $\hat{y}$ matching with y , i.e. with n data points,

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}(\hat{y}_i = y_i). \quad (12)$$

However, is it fool proof? Consider the case that we have $n = 100$ data points, only 1 data point is a positive class, say $y_{37} = 1$ and $y_i = 0$ for $i \neq 37$. Now consider the trivial predictor which predicts $\hat{y}_i = 0$ all i , the metric comes out to be

$$\frac{1}{100} \sum_{i=1}^{100} \mathbf{1}(\hat{y}_i = y_i) = 0.99. \quad (13)$$

Does this mean the predictor is 99% accurate? What if we are detecting for bombs at the airport?

Prediction and target relationship

There are 4 types of relationship between prediction and target.

- ▶ True Positives (TP) - the ones predicted to be positive and are indeed positives, $\hat{y} = 1$ and $y = 1$.
- ▶ True Negatives (TN) - the ones predicted to be negative and are indeed negative, $\hat{y} = 0$ and $y = 0$.
- ▶ False Positives (FP) - the ones predicted to be positive but are in fact negatives, $\hat{y} = 1$ and $y = 0$.
- ▶ False Negatives (FN) - the ones predicted to be negative and are indeed positives, $\hat{y} = 0$ and $y = 1$.

target \ prediction	$\hat{y} = 0$	$\hat{y} = 1$
$y = 0$	TN	FP
$y = 1$	FN	TP

Note that this table can be generalized for multi-class cases, and is known as confusion matrix.

Prediction and target relationship

Our naive metric earlier is essentially $TP + TN$. Often times, we are more interested in the FP and/or FN.

- ▶ When it is more costly to miss a positive case, e.g. bomb detection and COVID testing, FN should be emphasized.
- ▶ When it is more costly to miss a negative case, e.g. advertising and marketing, FN should be emphasized.

The four categories together provide the complete picture of the performance.

Additional metrics

Additional metrics can be derived from the four.

$$\text{precision} = \frac{TP}{TP + FP} \quad \text{recall} = \frac{TP}{TP + FN} \quad (14)$$

- ▶ Precision describes the proportion of “corrects” among the ones predicted positive.
- ▶ Recall is also known as True Positive Rate (TPR). It describes the proportion of total positives that can be retrieved, and is often used for various detection use cases.

Another common metric is False Positive Rate (FPR) is defined as $FP/(FP + TN)$.

F-score

As precision and recall are still two values, often we need to combine the two information into one score. This can be well summarized by F-score.

$$F_1 = \frac{2}{\text{precision}^{-1} + \text{recall}^{-1}} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (15)$$
$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}$$

F_1 score is the harmonic mean of precision and recall, and F_β adjusts the balance between precision and recall using β parameter.

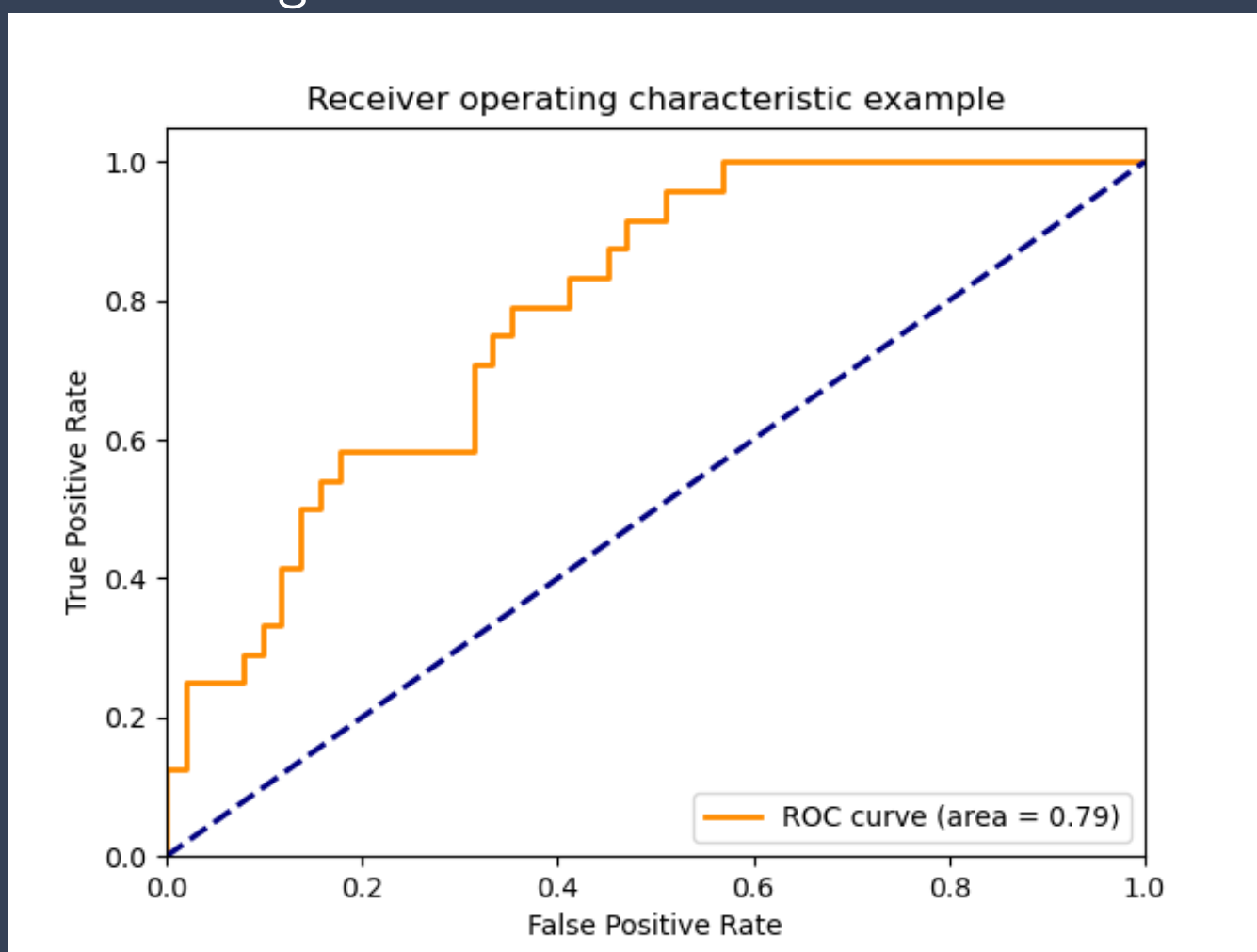
Area under the curve - ROC

We can also provide two plots to describe the performance based on soft labels. ROC (Receiver Operating Characteristic) curve

- ▶ plots TPR against FPR by computing these two metrics at possible thresholds;
- ▶ is monotonically increasing, as increasing FPR tolerance certainly increases TPR;
- ▶ the area under the curve is often used as a measure of the model performance.

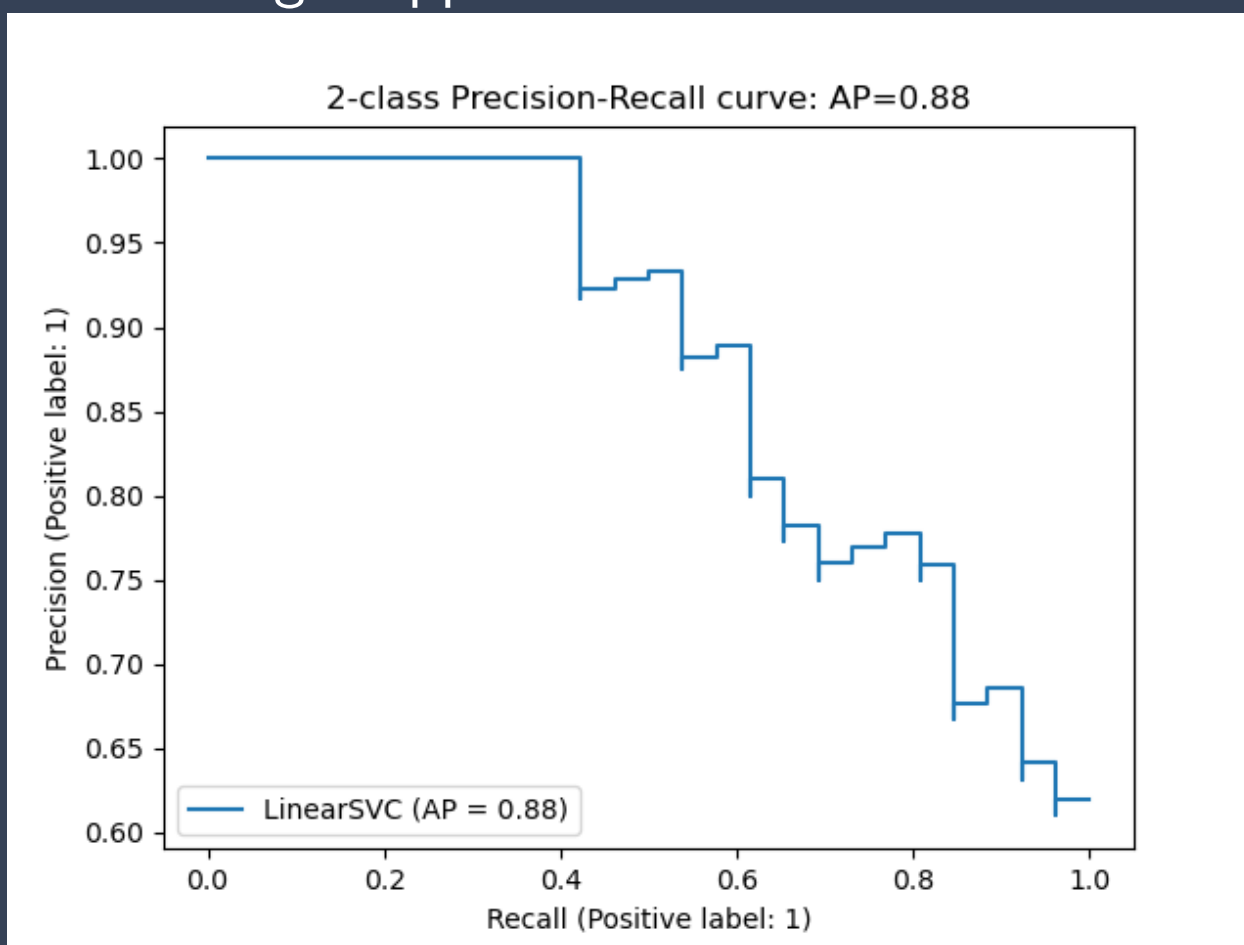
Area under the curve - ROC ²

Area under the ROC curve (AUC) is in range of $[0, 1]$ in theory. In practice, the range is $[0.5, 1]$, because a trivial predictor is capable of achieving AUC of 0.5.



Area under the curve - precision-recall ³

Precision-recall curve plots precision against recall. Likewise, the area under the curve is often used as a measure of the model performance. However, monotonicity and the lower bound of 0.5 are no longer applicable.



³Plot credit to scikit-learn

Continuous targets

Continuous targets

Continuous targets take value in $y \in \mathbb{R}$, and the prediction values also take from $\hat{y} \in \mathbb{R}$. Common metrics is a function of the difference of y and $\hat{y}$.

- Mean squared error (MSE)

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (16)$$

- Mean absolute deviation (MAD)

$$\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (17)$$